

Magnetic Earth Models

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Aims:

Within small groups, you will:

- Make a magnetic Earth model and a magnetic field detector
- Experiment using the field detector, a field compass, staples and/ or scourer pieces to discover what the magnetic field around your model looks like
- Draw a simple diagram of Earth's magnetic field based on your observations

Materials:

For the magnetic Earth model:

- Earth stress ball OR polystyrene sphere
- N-S bar magnet OR a pair of neodymium disk magnets
- Strong glue (if using disk magnets)

For the magnetic field detector:

- Pencil with eraser top
- Push pin
- Paper clip
- Flat nosed pliers and/ or wire cutters

Other equipment:

- Closed staples
- Stainless steel scourer (optional)
- Paper plate/ tray
- Scissors
- Field compass (optional)
- Pens/ pencils/ coloured pencils

Activity 1: Constructing the magnetic Earth models & field detector

Step 1: How to Assemble a Magnetic Earth:

- Ask a teacher to help you make the holes in your ball and insert the magnets.

Step 2: How to Assemble the Magnetic Field Detector:

Demonstration video: [Magnetic Earth Models Step 2: How to Assemble the Magnetic Field Detector](#)

- Straighten out a paper clip using pliers.
- Bend it into the shape in **Figure 1** and cut the ends until it is about 3 cm long.
- Place the end of a push pin through the loop in your paper clip and attach it to the eraser top on a pencil, as shown in **Figure 1**.
- Make sure your paperclip is free to spin between the pin and eraser.

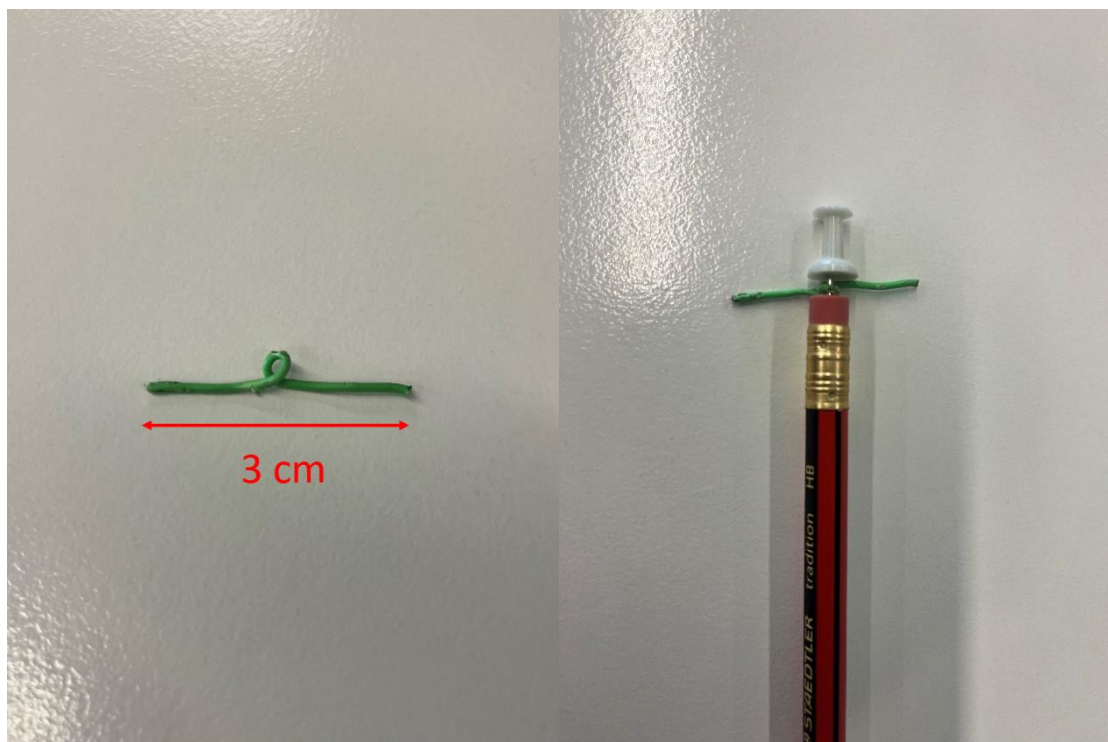


Figure 1: Paper clip bent into shape (left) and completed field detector (right).

Activity 2: Experimenting with the magnetic field around the magnetic Earth models

Imagine you are a scientist investigating the Earth's magnetic field.

Move your field detector around your model Earth and answer the following questions:

1. Where does the field detector point at the Earth's poles? How about the equator?
2. What happens when you move the field detector around the Earth?
3. Where do you think the magnetic field strength is the strongest – at the poles or the equator?

Your teacher may have given you a field compass. Move this around your Earth and see how the position of the compass needle changes.

Bonus question: Can you identify the north and south magnetic poles of your Earth?

HINT: The north end of the compass will be attracted to the south magnetic pole of your Earth.

Now we are going to visualise this magnetic field using closed staples and pieces of a stainless steel scourer. Play around with rolling/ pouring the staples and scourer pieces onto your Earth.

4. Where do the staples/ scourer pieces want to stick – at the centre of the Earth or the poles?
5. How does the orientation of the staples/ scourer pieces differ between the poles and the equator?

6. Can you get your staples to join up and form loops/ chains? Why do they do this?

Activity 3: Drawing the Earth's magnetic field

From your experimenting in activity 2, you should now have ideas about what the magnetic field around your model Earth looks like.

Your task is to draw what you think this field looks like.

The figure on the next page is your template. The rectangle inside of the 'Earth' represents the magnet(s) inside of your Earth model.

Add the following features to your diagram:

1. The magnetic north and south pole of the magnet
2. The geographic north and south poles of the Earth
3. Have a go at drawing lines to represent the magnetic field around the Earth – where do these start and end?
4. What direction are these field lines going – can you add arrows onto your lines?

Compare your drawing with a classmate from another group. Are your drawings similar?

Your teacher will now show you an actual diagram of Earth's magnetic field. How similar was your drawing?

